

ReportLab: Professional PDF Generation

What is ReportLab?

ReportLab is the industry standard for programmatic PDF generation in Python. It provides a powerful Platypus framework for automatic text flow and page breaks, professional typography control, and a low-level Canvas API for precise positioning.

Why Use ReportLab?

- **Platypus Framework** - Automatic text flow and page breaks
- **Professional Typography** - Kerning, leading, tracking control
- **Flowables** - Content blocks that position themselves
- **Stylesheets** - Separate content from formatting
- **Advanced Tables** - Spanning cells, conditional formatting
- **Canvas API** - Low-level control when needed
- **Production Proven** - Used by major companies worldwide
- **Pure Python** - No system dependencies
- **Small File Sizes** - Efficient PDF generation

How It Works

ReportLab uses an object-oriented approach where you build a 'story' of flowable objects (Paragraph, Table, Image, etc.) and then build the document. The Platypus framework automatically handles page breaks, text wrapping, and positioning.

Basic Example:

```
# Create document doc = SimpleDocTemplate("output.pdf", pagesize=letter) #
Get styles styles = getSampleStyleSheet() # Build story (content) story = []
story.append(Paragraph("Title", styles['Heading1'])) story.append(Spacer(1,
0.2*inch)) story.append(Paragraph("Body text...", styles['BodyText'])) #
Generate PDF doc.build(story)
```

Best Use Cases

- Professional documents with precise typography
- Data-heavy reports with complex tables

- Multi-page reports with automatic pagination
- Enterprise applications requiring production-grade PDFs
- Financial reports and legal documents
- Academic papers requiring precise formatting

Limitations & Considerations

- ■ **Learning Curve** - Medium complexity (1-2 days to master)
- ■ **Manual Markdown Parsing** - Must parse markdown yourself
- ■ **Limited CSS Support** - Uses ReportLab styles, not CSS
- ■ **More Code** - More verbose than HTML-based approaches
- ■ **No HTML Input** - Must convert HTML to ReportLab objects manually

Installation

```
pip install reportlab
```

No system dependencies required - pure Python!